Module 09 — Maven and Spring Boot Setup

Phase: Backend (first JSON API)
Previous module: JavaScript, DOM, and Forms (shared FRONTEND — also HTML · CSS and Responsive)
Next module: Spring Web REST

Module Outcome:

- Let Maven fetch jars so you are not hunting a driver on the desktop
- `mvn spring-boot:run` a Hello JSON endpoint on `:8080`
- Keep secrets in env / `.env.example` — not in `application.yml` on GitHub

Teach order today: classpath pain → Maven GAV + wrapper → Boot 3 parent + `starter-web` → Initializr / paste pom → `@SpringBootApplication` + `mvn spring-boot:run` → `application.yml` → `.env.example` → `@RestController` Hello + health JSON.

Prerequisites

- **JDK 17** (`java -version`, `javac -version`) and **Maven 3.9+** (`mvn -version`) — Tested Toolchain.
- Shared frontend complete enough to open JSON in a browser. `fetch` overview lives in JavaScript, DOM, and Forms — do not redo HTML / CSS / JS here.

Definition / Purpose

Definition: Maven is a **build tool**. You list libraries in **`pom.xml`**. Maven downloads them from Maven Central, compiles with JDK 17, and writes output under **`target/`** (Git week already ignores that folder).

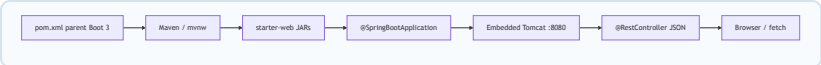
Definition: Spring Boot is an **opinionated way to start Spring**. A **starter** (here **`spring-boot-starter-web`**) is a curated set of JARs. **`@SpringBootApplication`** turns on auto-configuration. **`@RestController`** maps HTTP URLs to Java methods and writes **JSON**.

Purpose: JDBC week you typed `java -cp ".:mysql-connector-j-8.4.0.jar"`. One missing JAR and the run died. Teams cannot email JARs. Maven pins versions in `pom.xml`. Boot adds an **embedded Tomcat** so `mvn spring-boot:run` is a web server — Priya's browser and Ravi's `fetch` can hit `http://localhost:8080/hello`. **Later —** Spring Data JPA will sit on the JDBC habit you already have. Interviews still ask you to start a Boot app from a pom.

🏦 Bank 📦 Order 💻 Laptop 👤 Employee 📱 UPI

Where used?
- CampusCart / **TechMart** — first `/hello` and `/health` before catalog APIs
- **SecureBank** / UPI — a JSON health ping before any money movement
- **MedCare** — "is the API up?" before booking screens
- **AcmeHR** — a Boot service that later returns payslip JSON
- Ecommerce — the same `starter-web` shape for product APIs (**Later —** JPA)

Visual — Maven then Boot:



Key takeaways:

- Maven reads `pom.xml`; output is `target/` (never commit it)
- This course pins **Spring Boot 3.x** + **JDK 17** + `jakarta.*`
- Core path dependency: `spring-boot-starter-web` **only**
- First proof is `mvn spring-boot:run`, then **Hello JSON**

Motivation — Feel the pain of `-cp`

JDBC needed Connector/J on the command line. Two classmates, two slightly different JAR names — "it works on my laptop."

```
public class ClasspathPain {
    public static void main(String[] args) {
        System.out.println("java -cp .:mysql-connector-j-8.4.0.jar LibraryCrud");
        System.out.println("Ravi's folder has mysql-connector-j-8.4.1.jar");
        System.out.println("Maven: list the library once in pom.xml");
    }
}
```

Expected Output:

```
java -cp .:mysql-connector-j-8.4.0.jar LibraryCrud
Ravi's folder has mysql-connector-j-8.4.1.jar
Maven: list the library once in pom.xml
```

Remember: `pom.xml` is the packing list. `mvn` is the shopper.

Interview: "Maven vs `javac`?" — `javac` compiles one file. Maven compiles the module, pulls dependencies, and (with Boot) runs the app.

Important Concepts

1. Maven — GAV, `pom.xml`, wrapper

Definition: Every Maven artifact has **GAV**: **groupId**, **artifactId**, **version**. `pom.xml` (Project Object Model) is the XML file that lists your GAV and your **dependencies**. The **Maven Wrapper** (`mvnw` / `mvnw.cmd`) is a small script that downloads a pinned Maven so Priya without Maven on PATH can still run `./mvnw spring-boot:run`.

Purpose: Without GAV, "give me Spring Web" is vague. With GAV, Maven Central knows the exact JARs. Wrapper means the repo documents the Maven version — not each laptop.

Verify the toolchain:

```
java -version
javac -version
mvn -version
```

Expected Output: (vendor text varies; you want **17** and Maven **3.9+**)

```
openjdk version "17.x.x" ...
javac 17.x.x
Apache Maven 3.9.x ...
Java version: 17.x.x
```

If `mvn` is missing, install Maven 3.9+ (Tested Toolchain) or generate the project from start.spring.io so **`mvnw`** is included.

Note:
- Wrong: committing `target/` — Git week already ignores it; first `mvn` build must still show `target/` **untracked**.
- Wrong: treating Wrapper as optional on a team laptop that has no Maven — prefer `./mvnw` in README.
- `mvn` vs `./mvnw`: both run the same goals when Wrapper exists. Windows: `mvnw.cmd`.

Remember: `groupId` is the organisation (`in.ksr`). `artifactId` is the project folder name (`campus-hello`). Print the Maven GAV for campus-hello: groupId `in.ksr`, artifactId `campus-hello`, and version `0.0.1-SNAPSHOT`.

```
public class GavLabels {
    public static void main(String[] args) {
        System.out.println("groupId:    in.ksr");
        System.out.println("artifactId: campus-hello");
        System.out.println("version:    0.0.1-SNAPSHOT");
    }
}
```

Expected Output:

```
groupId:    in.ksr
artifactId: campus-hello
version:    0.0.1-SNAPSHOT
```

Print the three commands that start Spring Boot: installed `mvn`, macOS/Linux `./mvnw`, and Windows `mvnw.cmd`.

```
public class WrapperVsMvn {
    public static void main(String[] args) {
        System.out.println("Installed:  mvn spring-boot:run");
        System.out.println("macOS/Linux: ./mvnw spring-boot:run");
        System.out.println("Windows:     mvnw.cmd spring-boot:run");
    }
}
```

Expected Output:

```
Installed:  mvn spring-boot:run
macOS/Linux: ./mvnw spring-boot:run
Windows:     mvnw.cmd spring-boot:run
```

---

2. Spring Boot 3 parent, JDK 17, `starter-web` only

Definition: `spring-boot-starter-parent` is a Maven **parent POM**. It pins dependency versions so you rarely type a version on each starter. `java.version` **17** tells the compiler to use JDK 17. `spring-boot-starter-web` pulls Spring MVC, Jackson (JSON), and **embedded Tomcat**.

Purpose: Without the parent, you juggle twenty compatible versions. Without `starter-web`, there is no HTTP server. Extra starters this week are **out of window**.

Course pin: **Spring Boot 3.x** (examples use **3.5.16**) and `jakarta.*`. start.spring.io may default to **4.x** in 2026 — **pick 3.5.x** (Java 17) so this batch matches Tested Toolchain. If the site only offers 4.x, paste the parent version below by hand.

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>

    <parent>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-parent</artifactId>
        <version>3.5.16</version>
        <relativePath/>
    </parent>

    <groupId>in.ksr</groupId>
    <artifactId>campus-hello</artifactId>
    <version>0.0.1-SNAPSHOT</version>
    <name>campus-hello</name>

    <properties>
        <java.version>17</java.version>
    </properties>

    <dependencies>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-web</artifactId>
        </dependency>
    </dependencies>

    <build>
        <plugins>
            <plugin>
                <groupId>org.springframework.boot</groupId>
                <artifactId>spring-boot-maven-plugin</artifactId>
            </plugin>
        </plugins>
```

```
    </build>
</project>
```

Do not add this week: `spring-boot-starter-data-jpa`, `spring-boot-starter-security`, `spring-boot-starter-thymeleaf`, `spring-boot-starter-jdbc`, MySQL driver, Lombok. Those pull **Hikari / Security / templates / JPA** — all **Later —**.

JAR vs WAR (lite): default packaging is **JAR** — `java -jar target/campus-hello-0.0.1-SNAPSHOT.jar` runs with **embedded Tomcat**. **WAR** is for dropping a file into an **external** Tomcat (older office setup). This course ships **JAR**.

Note:
- Wrong: `javax.servlet` imports on Boot 3 — that is Boot 2. Use **`jakarta.*`** if you import servlet types.
- Wrong: `<java.version>1.8</java.version>` — class default is **17**.
- Parent manages versions; you still **declare** `starter-web` so Maven knows you want a web app.

Interview: Parent vs starter? Parent = version bill of materials + plugin defaults. Starter = the JARs you actually compile against.

Print which starter is allowed this week (`spring-boot-starter-web`) and which ones stay out: JPA, Security, Thymeleaf, and JDBC/Hikari.

```
public class AllowedStarter {
    public static void main(String[] args) {
        System.out.println("YES  spring-boot-starter-web");
        System.out.println("NO   starter-data-jpa / starter-security / thymeleaf");
        System.out.println("NO   starter-jdbc  (that pulls Hikari)");
    }
}
```

Expected Output:

```
YES  spring-boot-starter-web
NO   starter-data-jpa / starter-security / thymeleaf
NO   starter-jdbc  (that pulls Hikari)
```

Print one line each for JAR with embedded Tomcat, WAR on an external Tomcat, and that this course ships JAR.

```
public class JarVsWarLite {
    public static void main(String[] args) {
        System.out.println("JAR: java -jar app.jar  (embedded Tomcat)");
        System.out.println("WAR: copy into an external Tomcat  (older host)");
        System.out.println("This course: JAR");
    }
}
```

Expected Output:

```
JAR: java -jar app.jar  (embedded Tomcat)
WAR: copy into an external Tomcat  (older host)
This course: JAR
```

---

3. Initializr, folder tree, component scan

Definition: start.spring.io (Spring Initializr) generates a Maven tree: `pom.xml`, `mvnw`, `src/main/java`, `src/main/resources`. `@SpringBootApplication` includes **component scan** from **its own package downward**.

Purpose: A controller in the wrong package is invisible — Tomcat starts, `/hello` is 404. Scan starts at the application class package.

Initializr (or the same choices in IntelliJ):

Field	Value
Project	Maven
Language	Java
Spring Boot	**3.5.x** (not 4.x)
Group	`in.ksr`
Artifact	`campus-hello`
Packaging	**Jar**
Java	**17**
Dependencies	**Spring Web** only

Unzip. Tree you should recognise:

```
campus-hello/
  pom.xml
  mvnw
  mvnw.cmd
  .gitignore
  src/main/java/in/ksr/campushello/CampusHelloApplication.java
  src/main/resources/application.yml    (you add this)
  src/main/resources/application.properties  (Initializr may add this — prefer yml)
```

If the zip has `application.properties`, replace it with **`application.yml`** (or delete properties once yml exists — do not keep both for the same keys).

Note:
- Wrong: `HelloController` in `in.ksr.web` while the application class is `in.ksr.campushello` — **sibling packages are not scanned**.
- Prefer: `in.ksr.campushello.HelloController` (same package) or a **subpackage** `in.ksr.campushello.web`.
- College Wi-Fi blocking start.spring.io: paste the pom from the previous topic and create the two Java files by hand, then `mvn wrapper:wrapper` if you want `mvnw`.

Common Error: Opening the **parent** folder in the IDE so Maven cannot see `pom.xml`. Open **`campus-hello/`** (the folder that contains `pom.xml`).

Print a valid `CampusHelloApplication` package, two controller packages that component-scan will find, and the `in.ksr.web` 404 trap.

```
public class ComponentScanRule {
    public static void main(String[] args) {
        System.out.println("App class:  in.ksr.campushello.CampusHelloApplication");
        System.out.println("OK:         in.ksr.campushello.HelloController");
        System.out.println("OK:         in.ksr.campushello.web.HelloController");
        System.out.println("404 trap:   in.ksr.web.HelloController");
    }
}
```

Expected Output:

```
App class:  in.ksr.campushello.CampusHelloApplication
OK:         in.ksr.campushello.HelloController
OK:         in.ksr.campushello.web.HelloController
404 trap:   in.ksr.web.HelloController
```

Print the three Maven folders: `src/main/java`, `src/main/resources` for `application.yml`, and gitignored `target/`.

```
public class MavenSourceLayout {
    public static void main(String[] args) {
        System.out.println("src/main/java           Java sources");
        System.out.println("src/main/resources      application.yml");
        System.out.println("target/                 build output — gitignored");
    }
}
```

Expected Output:

```
src/main/java           Java sources
src/main/resources      application.yml
target/                 build output — gitignored
```

---

4. `@SpringBootApplication` and `mvn spring-boot:run`

Definition: `@SpringBootApplication` is a composite: `@Configuration` + `@EnableAutoConfiguration` + `@ComponentScan`. `SpringApplication.run` starts the JVM process, auto-configures Tomcat from `starter-web`, and listens on a port. `mvn spring-boot:run` (or `./mvnw spring-boot:run`) compiles and starts that `main`.

Purpose: You do not write a `public static void main` that constructs Tomcat by hand. Boot's `main` **is** the server.

Copy-paste valid (needs the Maven tree — not `javac` on a lone file):

```
package in.ksr.campushello;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class CampusHelloApplication {

    public static void main(String[] args) {
        SpringApplication.run(CampusHelloApplication.class, args);
```

```
    }
}
```

First proof (no controller yet):

```
cd campus-hello
mvn spring-boot:run
```

Look for a log line similar to **`Tomcat started on port 8080`**. Browser `http://localhost:8080/` may show a **Whitelabel** 404 — that still proves the server is up. JSON comes after you add a controller.

Note:
- Wrong: `javac CampusHelloApplication.java` without Maven — `org.springframework` is not on a bare JDK classpath.
- Port **8080 in use** → set `server.port` in `application.yml` (next topic) or stop the other process.
- Stop the run with **Ctrl+C** in that terminal.

Interview: Why embedded Tomcat? So one JAR (or `spring-boot:run`) is enough in class and on many PaaS hosts. You are not installing a separate Tomcat to learn Hello JSON.

Print the three annotations inside `@SpringBootApplication`: `@Configuration`, `@EnableAutoConfiguration`, and `@ComponentScan`.

```
public class SpringBootApplicationParts {
    public static void main(String[] args) {
        System.out.println("@Configuration");
        System.out.println("@EnableAutoConfiguration");
        System.out.println("@ComponentScan");
        System.out.println("Together: @SpringBootApplication");
    }
}
```

Expected Output:

```
@Configuration
@EnableAutoConfiguration
@ComponentScan
Together: @SpringBootApplication
```

Print the `mvn spring-boot:run` command, the Tomcat-started-on-8080 log line to look for, and that a 404 on `/` is OK before `@RestController`.

```
public class RunProofLog {
    public static void main(String[] args) {
        System.out.println("mvn spring-boot:run");
        System.out.println("Look for: Tomcat started on port 8080");
        System.out.println("404 on / is OK before @RestController");
    }
}
```

Expected Output:

```
mvn spring-boot:run
Look for: Tomcat started on port 8080
404 on / is OK before @RestController
```

---

5. `application.yml` — port, name, placeholders

Definition: `application.yml` is Spring Boot's **config file** under `src/main/resources`. YAML uses indentation (spaces, not tabs). `${ENV:default}` reads an environment variable with a fallback.

Purpose: Hard-coding `8080` in Java means a rebuild to change port. YAML (and env) change behaviour without touching controllers. JDBC week had **no** `application.yml` — that file is **Spring**.

```
spring:
  application:
    name: campus-hello

server:
  port: ${SERVER_PORT:8080}

app:
  message: ${APP_MESSAGE:Hello, CampusCart}
```

Spring maps `SERVER_PORT` to `server.port`. `APP_MESSAGE` is your own key — the controller can read `app.message` with `@Value` (shown with the controller).

Note:
- Wrong: mixing `application.properties` and `application.yml` for the same key — pick **yml** in this course.
- Wrong: putting a MySQL password in yml and pushing it — there is **no database** this module; still never commit secrets.
- Indentation matters. `server:` then two spaces then `port:`.

Common Error: Tab characters in YAML → Boot fails at start with a parse error. Use spaces.

Remember: Change port in yml, restart `spring-boot:run`. Config is not hot-reloaded unless you add extra tools (**Later —**).

Print the three YAML keys for CampusCart: `spring.application.name`, `server.port` with a `SERVER_PORT` default, and `app.message` with an `APP_MESSAGE` default.

```
public class YmlKeysPrint {
    public static void main(String[] args) {
        System.out.println("spring.application.name=campus-hello");
        System.out.println("server.port=${SERVER_PORT:8080}");
        System.out.println("app.message=${APP_MESSAGE:Hello, CampusCart}");
    }
}
```

Expected Output:

```
spring.application.name=campus-hello
server.port=${SERVER_PORT:8080}
app.message=${APP_MESSAGE:Hello, CampusCart}
```

Print what to set when port 8080 is busy (`server.port: 8081`) and the URL to open for `/hello`.

```
public class PortBusyHint {
    public static void main(String[] args) {
        System.out.println("If 8080 is busy: server.port: 8081");
        System.out.println("Then open http://localhost:8081/hello");
    }
}
```

Expected Output:

```
If 8080 is busy: server.port: 8081
Then open http://localhost:8081/hello
```

---

6. Secrets — `.env.example` (still no DataSource)

Definition: A **secret** is a password or token that must not appear in git. Git week ignores `.env`. You commit `.env.example` with dummy keys. Spring Boot **does not** load a `.env` file by itself — it reads **environment variables**. Copy `.env.example` → `.env` on your machine, then **export** keys or use your IDE env dialog.

Purpose: Next modules will add DB passwords. The habit starts **before** JPA: dummy keys in git, real values local. This week there is **no** `MYSQL_PASSWORD` required — optional `APP_MESSAGE` / `SERVER_PORT` only.

`.env.example` (commit this):

```
# Copy to .env — never commit .env
# Spring Boot reads OS environment variables (not this file automatically).
SERVER_PORT=8080
APP_MESSAGE=Hello from Priya laptop
```

Export then run (macOS / Linux):

```
export SERVER_PORT=8080
export APP_MESSAGE="Hello from Priya laptop"
mvn spring-boot:run
```

Windows PowerShell: `$env:SERVER_PORT=8080` then `mvn spring-boot:run`.

Note:
- Wrong: committing `.env` “because the repo is private.”
- Wrong: adding a `mysql` password **and** `spring.datasource.*` this week — that is **Later —** JPA, and it pulls a pool.
- Prefer: `.env.example` in git; real `.env` local; README says “export or IDE env.”

Remember: Env first. Never a password literal in `application.yml` on GitHub.

Print `app.message` when the env var is missing so the YAML default `Hello, CampusCart` is used.

```
public class EnvOrDefault {
    public static void main(String[] args) {
        String message = System.getenv("APP_MESSAGE");
        if (message == null || message.length() == 0) {
            message = "Hello, CampusCart";
        }
        System.out.println(message);
    }
}
```

Expected Output: (when `APP_MESSAGE` is unset)

```
Hello, CampusCart
```

Print git rules for `.env.example` vs `.env`, and that Boot reads OS env — not a `.env` file by itself.

```
public class EnvGitRule {
    public static void main(String[] args) {
        System.out.println("git: .env.example  YES");
        System.out.println("git: .env          NO");
        System.out.println("Boot reads OS env, not the .env file by itself");
    }
}
```

Expected Output:

```
git: .env.example  YES
git: .env          NO
Boot reads OS env, not the .env file by itself
```

---

7. `@RestController` + `@GetMapping` Hello JSON

Definition: `@RestController` is `@Controller` + `@ResponseBody`: every method return value is the **HTTP body**, not a template name. `@GetMapping("/hello")` handles HTTP **GET** `/hello`. Returning a `Map` (or a small type with getters) is serialised to **JSON** by Jackson (inside `starter-web`).

Purpose: JS week called `fetch` on jsonplaceholder. Today **your** laptop is the API. **Later —** `@PathVariable`, `@RequestBody`, status codes (next module). **Later —** `@Controller` returning a Thymeleaf view name.

Copy-paste valid (same package as the application class):

```
package in.ksr.campushello;

import java.util.Map;
import org.springframework.beans.factory.annotation.Value;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class HelloController {
```

```
    @Value("${app.message:Hello, CampusCart}")
    private String message;

    @GetMapping("/hello")
    public Map<String, String> hello() {
        return Map.of("message", message);
    }

    @GetMapping("/health")
    public Map<String, String> health() {
        return Map.of("status", "UP", "app", "campus-hello");
    }
}
```

Restart `mvn spring-boot:run`. Browser or:

```
curl http://localhost:8080/hello
curl http://localhost:8080/health
```

Expected Output: (default yml)

```
{"message":"Hello, CampusCart"}
```

```
{"status":"UP","app":"campus-hello"}
```

`Map.of` key order is not guaranteed — both keys present is success.

`jakarta.*`: Boot 3 servlet types live in **`jakarta.servlet`**, not `javax.servlet`. A Hello `Map` does not need that import. If you later take `HttpServletRequest`, write `import jakarta.servlet.http.HttpServletRequest;` — **`javax.*` will not compile** on Boot 3.

Note:
- Wrong: `@Controller` without `@ResponseBody` for JSON — Boot looks for a **view** (**Later —** Thymeleaf) and you get an error, not JSON.
- Wrong: `GET /Hello` vs `/hello` — paths are **case-sensitive**.
- Do **not** add Actuator for `/health` this week. A **plain** **`@GetMapping("/health")`** is the assignment. **Later —** actuator in deploy lab.

Interview: `@RestController` vs `@Controller`? REST + JSON vs (typically) HTML templates. This module is JSON-only.

Print the two GET paths (`/hello` and `/health`) and that `@RestController` returns a JSON body, not an HTML file.

```
public class MappingNamesPrint {
    public static void main(String[] args) {
        System.out.println("GET /hello  -> JSON message");
        System.out.println("GET /health -> JSON status UP");
        System.out.println("@RestController = JSON body, not a .html file");
    }
}
```

Expected Output:

```
GET /hello  -> JSON message
GET /health -> JSON status UP
@RestController = JSON body, not a .html file
```

Print Boot 2 `javax.servlet` vs Boot 3 `jakarta.servlet`, and that the wrong import fails with cannot find symbol.

```
public class JakartaNotJavax {
    public static void main(String[] args) {
        System.out.println("Boot 2 / old Java EE: javax.servlet");
        System.out.println("Boot 3 / Jakarta EE:  jakarta.servlet");
        System.out.println("Wrong import = cannot find symbol");
    }
}
```

Expected Output:

```
Boot 2 / old Java EE: javax.servlet
Boot 3 / Jakarta EE:  jakarta.servlet
Wrong import = cannot find symbol
```

JS already knows `fetch` (FRONTEND/03). After Hello runs:

```
fetch("http://localhost:8080/hello")
  .then((res) => res.json())
  .then((data) => console.log(data.message));
```

That is a **browser console** check — do not rewrite the JS chapter. **Later —** CORS when a Vite app on :5173 calls :8080.

Step-by-Step Annotated Java Code

Generate **`campus-hello`** (Initializr **or** paste `pom.xml`). JDK 17. **Spring Web** only. Open the folder that contains `pom.xml`.

Step 1: Prove Maven and JDK

```
java -version
javac -version
mvn -version
```

Print the three PATH tools you need: `java`, `javac` (Maven still uses it), and `mvn` to read `pom.xml`.

```
public class PathTools {
    public static void main(String[] args) {
        System.out.println("java   - run bytecode");
        System.out.println("javac  - still used by Maven");
        System.out.println("mvn    - reads pom.xml");
```

```
    }
}
```

Expected Output:

```
java   — run bytecode
javac  — still used by Maven
mvn    — reads pom.xml
```

Print that `mvn -version` must show Java 17, and that Java 8 or 11 means you must fix `JAVA_HOME`.

```
public class JavaHomeHint {
    public static void main(String[] args) {
        System.out.println("mvn -version must show Java version: 17");
        System.out.println("If it shows 8 or 11, fix JAVA_HOME");
    }
}
```

Expected Output:

```
mvn -version must show Java version: 17
If it shows 8 or 11, fix JAVA_HOME
```

Step 2: Application class in the right package

Keep **`CampusHelloApplication`** where Initializr put it (`in.ksr.campushello`). Do not move it to the default package.

Print the `SpringApplication.run(CampusHelloApplication.class, args)` call and that this line starts the server.

```
public class BootMainSketch {
    public static void main(String[] args) {
        System.out.println("SpringApplication.run(CampusHelloApplication.class, args)");
        System.out.println("That line is the server start");
    }
}
```

Expected Output:

```
SpringApplication.run(CampusHelloApplication.class, args)
That line is the server start
```

Print the default-package trap: a Boot class with no `package` confuses component scan — always use `in.ksr.campushello`.

```
public class DefaultPackageTrap {
    public static void main(String[] args) {
        System.out.println("No package { } on the Boot class");
        System.out.println("Component scan gets confused — always use in.ksr.campushello");
```

```
    }
}
```

Expected Output:

```
No package { } on the Boot class
Component scan gets confused — always use in.ksr.campushello
```

Step 3: First `mvn spring-boot:run`

```
mvn spring-boot:run
```

Wait until Tomcat started. Leave it running. Open a **second** terminal for `curl`.

Print the two success log tokens: `Started CampusHelloApplication` and `Tomcat started on port 8080 (http)`.

```
public class TomcatStartedTokens {
    public static void main(String[] args) {
        System.out.println("Started CampusHelloApplication");
        System.out.println("Tomcat started on port 8080 (http)");
    }
}
```

Expected Output:

```
Started CampusHelloApplication
Tomcat started on port 8080 (http)
```

Print that the first `mvn` run needs Maven Central (network), then JARs sit in the local `.m2` cache.

```
public class FirstRunNetwork {
    public static void main(String[] args) {
        System.out.println("First mvn run needs Maven Central (network)");
        System.out.println("Then JARs sit in the local .m2 cache");
    }
}
```

Expected Output:

```
First mvn run needs Maven Central (network)
Then JARs sit in the local .m2 cache
```

Step 4: Add `application.yml` and `.env.example`

Create `src/main/resources/application.yml` (keys from concept 5). Add `.env.example` at the project root. Confirm `.gitignore` still has `target/` and `.env` (Git kit).

Restart Boot after yml changes.

Print that `application.yml` lives at `src/main/resources/application.yml`, not next to `pom.xml`.

```
public class ResourcesPathPrint {
    public static void main(String[] args) {
        System.out.println("src/main/resources/application.yml");
        System.out.println("Not next to pom.xml");
    }
}
```

Expected Output:

```
src/main/resources/application.yml
Not next to pom.xml
```

Print that changing `server.port` requires Ctrl+C and `mvn spring-boot:run` again.

```
public class RestartAfterYml {
    public static void main(String[] args) {
        System.out.println("Change server.port");
        System.out.println("Ctrl+C then mvn spring-boot:run again");
    }
}
```

Expected Output:

```
Change server.port
Ctrl+C then mvn spring-boot:run again
```

Step 5: `HelloController` — `/hello` and `/health`

Add `HelloController.java` in **`in.ksr.campushello`**. Restart. Then:

```
curl http://localhost:8080/hello
curl http://localhost:8080/health
```

Print the two URLs to hit: `http://localhost:8080/hello` and `http://localhost:8080/health`.

```
public class TwoGetPaths {
    public static void main(String[] args) {
        System.out.println("http://localhost:8080/hello");
        System.out.println("http://localhost:8080/health");
    }
}
```

Expected Output:

```
http://localhost:8080/hello
http://localhost:8080/health
```

Print that `/hello` is not an HTML page, then print the JSON body `{"message":"Hello, CampusCart"}`.

```
public class JsonIsTheBody {
    public static void main(String[] args) {
        System.out.println("Not an HTML page");
        System.out.println("{\"message\":\"Hello, CampusCart\"}");
    }
}
```

Expected Output:

```
Not an HTML page
{"message":"Hello, CampusCart"}
```

Step 6: Git hygiene before you push

`git status` must **not** list `target/` or `.env`. README: JDK 17, Boot 3, `mvn spring-boot:run`, two curl lines. See Standard Project Deliverables.

Print what to ignore (`target/`, `.env`) and what to keep (`pom.xml`, `mvnw`, `src/`, `.env.example`).

```
public class BootGitignore {
    public static void main(String[] args) {
        System.out.println("ignore: target/");
        System.out.println("ignore: .env");
        System.out.println("keep:   pom.xml mvnw src/ .env.example");
    }
}
```

Expected Output:

```
ignore: target/
ignore: .env
keep:   pom.xml mvnw src/ .env.example
```

Print the two README run lines: `mvn spring-boot:run` and `curl http://localhost:8080/hello`.

```
public class ReadmeRunLines {
    public static void main(String[] args) {
        System.out.println("mvn spring-boot:run");
        System.out.println("curl http://localhost:8080/hello");
    }
}
```

Expected Output:

```
mvn spring-boot:run
curl http://localhost:8080/hello
```

Practice Exercises

(Each Practice item includes a filled Model answer: and Expected Output: where the demo prints.)

Basic

1. Print the GAV for a canteen hello app (`in.ksr` / `canteen-hello` / `0.0.1-SNAPSHOT`).

Model answer:

```
public class CanteenGav {
    public static void main(String[] args) {
        System.out.println("in.ksr");
        System.out.println("canteen-hello");
        System.out.println("0.0.1-SNAPSHOT");
    }
}
```

Expected Output:

```
in.ksr
canteen-hello
0.0.1-SNAPSHOT
```

2. List starters that are **banned** on this module's core path.

Model answer:

```
public class BannedStarters {
    public static void main(String[] args) {
        System.out.println("data-jpa");
        System.out.println("security");
        System.out.println("thymeleaf");
        System.out.println("jdbc / hikari");
        System.out.println("lombok (not required)");
    }
}
```

Expected Output:

```
data-jpa
security
thymeleaf
jdbc / hikari
lombok (not required)
```

3. Print the two JSON paths this module must expose.

Model answer:

```
public class RequiredJsonPaths {
    public static void main(String[] args) {
        System.out.println("GET /hello");
        System.out.println("GET /health");
    }
}
```

Expected Output:

```
GET /hello
GET /health
```

Intermediate

1. Sketch `application.yml` keys for name `upi-ping` and port `8081` (print the tree).

Model answer:

```
public class YmlUpiPing {
    public static void main(String[] args) {
        System.out.println("spring.application.name: upi-ping");
        System.out.println("server.port: 8081");
    }
}
```

Expected Output:

```
spring.application.name: upi-ping
server.port: 8081
```

2. Why would `/hello` 404 after a successful `spring-boot:run`?

Model answer:

```
public class Hello404Causes {
    public static void main(String[] args) {
        System.out.println("Controller missing @RestController or @GetMapping");
        System.out.println("Controller in a sibling package (not scanned)");
        System.out.println("Wrong URL /Hello vs /hello");
    }
}
```

Expected Output:

```
Controller missing @RestController or @GetMapping
Controller in a sibling package (not scanned)
Wrong URL /Hello vs /hello
```

3. Print a safe `.env.example` for this week (no MySQL password required).

Model answer:

```
public class EnvExampleThisWeek {
    public static void main(String[] args) {
        System.out.println("SERVER_PORT=8080");
        System.out.println("APP_MESSAGE=Hello from Ravi laptop");
    }
}
```

Expected Output:

```
SERVER_PORT=8080
APP_MESSAGE=Hello from Ravi laptop
```

Interview-Oriented

1. Maven vs Spring Boot?

Model answer: Maven is the **build tool** (`pom.xml`, dependencies, `target/`). Spring Boot is the **framework** that auto-configures a web app. Boot projects **use** Maven (this course) or Gradle.

2. What does `spring-boot-starter-web` add?

Model answer: Spring MVC, Jackson JSON, and **embedded Tomcat**. Enough for `@RestController` JSON. It does **not** add JPA, Security, or Thymeleaf.

3. `@SpringBootApplication` in one sentence?

Model answer: It marks the entry class and turns on configuration, auto-configuration, and component scan from that package down.

4. Why `application.yml` and not JDBC-style env-only for the port?

Model answer: Boot's native config file is `application.yml` (or properties). Env vars can **override** yml (`SERVER_PORT`). JDBC week had no Boot, so no yml.

5. Why `jakarta.*` on Boot 3?

Model answer: Java EE moved to Jakarta EE. Package names changed from `javax.*` to `jakarta.*`. Boot 3 follows Jakarta. Copy-pasting Boot 2 servlet imports fails to compile.

6. JAR vs WAR for this course?

Model answer: JAR with embedded Tomcat (`java -jar` / `spring-boot:run`). WAR is for an external servlet container — not the class default.

Mini-Project (Assignment): CampusCart / TechMart Hello Boot API

There is **no lab-kit folder** this module — generate **`campus-hello`** from start.spring.io (Maven, Java 17, Boot **3.5.x**, **Spring Web** only) or paste the `pom.xml` from this chapter. Push a public repo. No secrets in git.

This is the **first Spring artefact** on your GitHub: a Boot 3 app that returns Hello JSON. JDBC stays a separate CLI. Do **not** add JPA, Security, Thymeleaf, Lombok, or a DataSource to pass.

What to build

Piece	Detail
`pom.xml`	Parent **Spring Boot 3.x**, `<java.version>17</java.version>`, **`spring-boot-starter-web` only**, `spring-boot-maven-plugin`
Wrapper	`mvnw` from Initializr **or** `mvn spring-boot:run` documented if Wrapper is missing
`CampusHelloApplication`	`@SpringBootApplication` + `SpringApplication.run`

Piece	Detail
`HelloController`	`@RestController`; **GET** `/hello` JSON; **GET** `/health` JSON (`status` = `UP`)
`application.yml`	`spring.application.name`, `server.port` (placeholder OK)
`.env.example`	Dummy `SERVER_PORT` / `APP_MESSAGE` — **no** real passwords
`.gitignore`	`target/`, `.env`, `*.class`, `.idea/` (Git week)
README	JDK 17, Boot 3, run command, two curl examples
Demo `main` path	`mvn spring-boot:run` (or `./mvnw spring-boot:run`) until Tomcat started

Sample curl (after run):

```
{"message":"Hello, CampusCart"}
```

```
{"status":"UP","app":"campus-hello"}
```

Starter idea (controller body only — place it under `src/main/java/in/ksr/campushello/`):

```
package in.ksr.campushello;

import java.util.Map;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class HelloControllerLite {

    @GetMapping("/hello")
    public Map<String, String> hello() {
        return Map.of("message", "Hello, CampusCart");
    }

    @GetMapping("/health")
    public Map<String, String> health() {
        return Map.of("status", "UP");
    }
}
```

Wire this next to `CampusHelloApplication`. Do not `javac` this file alone.

Stretch

- `@Value("${app.message:Hello, CampusCart}")` so YAML / `APP_MESSAGE` changes the JSON.
- Extra GET `/canteen` returning `Map.of("stall", "South Indian", "item", "Idli")`.
- Screenshot of browser JSON + `mvn -version` output in README.
- Lombok `@Getter` on a tiny type — **optional**, not in Done when. Prefer `Map` or explicit getters.

- **Later —** `@PathVariable`, `@RequestBody`, status codes, Thymeleaf one-pager (next module). **Later —** JPA + MySQL.

Done when

- [ ] `mvn spring-boot:run` or `./mvnw spring-boot:run` starts (Tomcat log on the configured port)
- [ ] GET `/hello` returns JSON with a message
- [ ] GET `/health` returns JSON with `status` UP
- [ ] `application.yml` exists under `src/main/resources`
- [ ] `.env.example` is in git; `.env` is **not**
- [ ] `pom.xml` parent is Spring Boot **3.x**, Java **17**, **no** JPA / Security / Thymeleaf / `starter-jdbc` / Hikari as required
- [ ] No Lombok required to pass
- [ ] `target/` untracked; short README with run + curl (Standard Project Deliverables)

Interview Q&A

Shared with students after concepts and the mini-project. Use for review and interview prep — questions and model answers are included.

Q1. What is Maven?
A: A Java build tool. `pom.xml` lists GAV and dependencies. Maven downloads JARs, compiles, and writes `target/`. Wrapper (`mvnw`) pins Maven for classmates.

Q2. What is a GAV?
A: groupId + artifactId + version — the coordinates Maven Central uses. Example: `in.ksr` / `campus-hello` / `0.0.1-SNAPSHOT`.

Q3. What is Spring Boot?
A: A way to start Spring with auto-configuration and starters. `spring-boot-starter-web` gives you an embedded Tomcat and JSON MVC so `mvn spring-boot:run` is a web server.

Q4. Parent POM vs starter?
A: The **parent** (`spring-boot-starter-parent`) manages versions and plugin defaults. A **starter** is the dependency you declare (`starter-web`). You need both in this course's pom.

Q5. Why JDK 17 in `java.version`?
A: Course / hiring default (LTS). Boot 3 supports 17. Mixing 8/11 as `JAVA_HOME` while the pom says 17 causes confusing compiler errors.

Q6. What does `@SpringBootApplication` do?
A: Combines `@Configuration`, `@EnableAutoConfiguration`, and `@ComponentScan`. Scan starts at that class's package.

Q7. Why did `/hello` 404 even though Tomcat started?
A: No matching `@GetMapping`, or the controller was **outside** the scanned package, or the URL does not match (case / extra path).

Q8. `@RestController` vs `@Controller`?
A: `@RestController` writes the method return value as the body (JSON with Jackson). `@Controller` typically returns a **view** name. HTML templates are **Later —** Thymeleaf, not this module.

Q9. Why `application.yml`?

A: Boot's config file for port, app name, and placeholders. JDBC did not use it because JDBC was not Spring. Do not put secrets in yml on git.

Q10. Does Boot load `.env` automatically?

A: No. It reads **environment variables** (and yml). Commit `.env.example`; export keys or set them in the IDE. Never commit `.env`.

Q11. `javax.*` vs `jakarta.*`?

A: Boot 3 uses Jakarta (`jakarta.servlet…`). Boot 2 used `javax.*`. Wrong import: cannot find symbol.

Q12. JAR or WAR?

A: This course: **JAR** + embedded Tomcat. WAR is for deploying to an external Tomcat — mention in interviews, do not require it here.

Q13. Why not JPA / Security / Thymeleaf this week?

A: Teaching order: run Boot, return JSON, then REST mappings, then ORM and auth. Extra starters hide that path and pull pools or login filters you have not been taught.

Q14. Maven vs Gradle?

A: Both build Java. This course uses **Maven** (`pom.xml`) because hiring still lists it heavily and Initializr Maven is the class default. Gradle is another build tool — not required here.

Q15. How does this connect to `fetch`?

A: JS `fetch(url).then(r => r.json())` needs a URL that returns JSON. `/hello` is that URL on localhost. CORS for a separate React origin is **Later —**.

Next

Push **Campus Hello Boot API** (no secrets, `target/` ignored). Continue with **Spring Web REST** — more mappings, JSON, status codes, `@PathVariable` / `@RequestBody`, and a **Thymeleaf overview only**. JPA stays after that.